



LLaMA Factory: Unified Efficient Fine-Tuning of 100+ LLMs

Visit [GitHub Page](#) [Documentation](#) [Blog](#)

Language en ▼	Model name Input the initial name to search for the model. Qwen3-1.7B-Base ▼	Model path Path to pretrained model or model identifier from Hugging Face. Qwen/Qwen3-1.7B-Base
Hub name Choose the model download source. huggingface ▼		
Finetuning method lora ▼	Checkpoint path ▼	
Quantization bit Enable quantization (QLoRA). none ▼	Quantization method Quantization algorithm to use. bnb ▼	Chat template The chat template used in constructing prompts. default ▼
RoPE scaling RoPE scaling method to use. none ▼	Booster Approach used to boost training speed. auto ▼	
Train Evaluate & Predict Chat Export		
Stage The stage to perform in training. Supervised Fine-Tune	Data dir Path to the data directory. data	Dataset support_tickets × × ▼

Preview dataset

Learning rate

Initial learning rate for AdamW.

5e-5

Epochs

Total number of training epochs to perform.

3.0

Maximum gradient norm

Norm for gradient clipping.

1.0

Seed

Random seed for training.

42

Max samples

Maximum samples per dataset.

100000

Compute type

Whether to use mixed precision training.

bf16

Cutoff length

Max tokens in input sequence.

2048

131072

4

Batch size

Number of samples processed on each GPU.

2

1024

1

Gradient accumulation

Number of steps for gradient accumulation.

8

1024

1

Val size

Percentage of validation set from the entire dataset.

0

1

0

LR scheduler

Name of the learning rate scheduler.

cosine

Extra configurations

Logging steps

Number of steps between two logs.

5

1

1000

Save steps

Number of steps between two checkpoints.

100

10

5000

Warmup steps

Number of steps used for warmup.

0

0

5000

NEFTune alpha

Magnitude of noise adding to embedding vectors.

0

10

0

Extra arguments

Extra arguments passed to the trainer in JSON format.

{"optim": "adamw_torch"}

☐ Pack sequences

Pack sequences into samples of fixed length.

☐ Use neat packing

Avoid cross-attention between packed sequences.

☐ **Train on prompt**

Disable the label mask on the prompt (only for SFT).

☐ **Mask history**

Train on the last turn only (only for SFT).

☐ **Resize token embeddings**

Resize the tokenizer vocab and the embedding layers.

☐ **Enable LLaMA Pro**

Make the parameters in the expanded blocks trainable.

☐ **Enable thinking**

Whether or not to enable thinking mode for reasoning models.

Enable external logger

Use TensorBoard or wandb to log experiment.

none

Trackio Settings



Freeze tuning configurations



LoRA configurations



RLHF configurations



Multimodal configurations



GaLore configurations



APOLLO configurations



BAdam configurations



SwanLab configurations



Preview command

Save arguments

Load arguments

Abort

Output dir Directory for saving results.	Config path Path to config saving arguments.
train_2026-09-12-13-56-28 ▼	2026-09-12-13-56-28.yaml ▼

Device count Number of devices available.	DeepSpeed stage DeepSpeed stage for distributed training.	☐ Enable offload Enable DeepSpeed offload (slow down training).
1	none ▼	

Finished.

[illegible]


```

        "full_attention",
        "full_attention",
        "full_attention",
        "full_attention",
        "full_attention",
        "full_attention",
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    "max_window_layers": 28,
    "model_type": "qwen3",
    "num_attention_heads": 16,
    "num_hidden_layers": 28,
    "num_key_value_heads": 8,
    "pad_token_id": null,
    "rms_norm_eps": 1e-06,
    "rope_parameters": {
        "rope_theta": 1000000,
        "rope_type": "default"
    },
    "sliding_window": null,
    "tie_word_embeddings": true,
    "transformers_version": "5.8.0",
    "use_cache": true,
    "use_sliding_window": false,
    "vocab_size": 151936
}

```

```

[INFO|2026-09-12 14:01:40] configuration_utils.py:780 >> loading configuration
[INFO|2026-09-12 14:01:40] configuration_utils.py:856 >> Model config Qwen3Con

```

```

    "architectures": [
        "Qwen3ForCausalLM"
    ],
    "attention_bias": false,
    "attention_dropout": 0.0,
    "bos_token_id": 151643,
    "dtype": "bfloat16",
    "eos_token_id": 151643,
    "head_dim": 128,
    "hidden_act": "silu",
    "hidden_size": 2048,
    "initializer_range": 0.02,
    "intermediate_size": 6144,
    "layer_types": [
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        "full_attention",
        "full_attention",
        "full_attention",
        "full_attention",
        "full_attention",
    ],

```



```

    "output_hidden_states": false,
    "use_cache": false
}

```

```

[INFO|2026-09-12 14:09:55] configuration_utils.py:1039 >> loading configuratio
[INFO|2026-09-12 14:09:55] configuration_utils.py:1086 >> Generate config Gene
    "bos_token_id": 151643,
    "do_sample": false,
    "eos_token_id": 151643,
    "max_new_tokens": 2048
}

```

```

[INFO|2026-09-12 14:09:55] dynamic_module_utils.py:441 >> Could not locate the
[INFO|2026-09-12 14:09:55] logging.py:144 >> Gradient checkpointing enabled.
[INFO|2026-09-12 14:09:55] logging.py:144 >> Using torch SDPA for faster train
[INFO|2026-09-12 14:09:55] logging.py:144 >> Upcasting trainable params to flo
[INFO|2026-09-12 14:09:55] logging.py:144 >> Fine-tuning method: LoRA
[INFO|2026-09-12 14:09:55] logging.py:144 >> Found linear modules: gate_proj,k
[INFO|2026-09-12 14:09:57] logging.py:144 >> trainable params: 8,716,288 || al
[WARNING|2026-09-12 14:09:57] trainer_utils.py:1238 >> The tokenizer has new P
[INFO|2026-09-12 14:09:57] trainer.py:1469 >> ***** Running training *****
[INFO|2026-09-12 14:09:57] trainer.py:1470 >>     Num examples = 467
[INFO|2026-09-12 14:09:57] trainer.py:1471 >>     Num Epochs = 3
[INFO|2026-09-12 14:09:57] trainer.py:1472 >>     Num update steps per epoch = 3
[INFO|2026-09-12 14:09:57] trainer.py:1473 >>     Instantaneous batch size per d
[INFO|2026-09-12 14:09:57] trainer.py:1476 >>     Total train batch size (w. par
[INFO|2026-09-12 14:09:57] trainer.py:1477 >>     Gradient Accumulation steps =
[INFO|2026-09-12 14:09:57] trainer.py:1478 >>     Total optimization steps = 90
[INFO|2026-09-12 14:09:57] trainer.py:1479 >>     Number of trainable parameters
[INFO|2026-09-12 14:10:35] logging.py:144 >> {'loss': 7.0932, 'learning_rate':
[INFO|2026-09-12 14:11:08] logging.py:144 >> {'loss': 3.4539, 'learning_rate':
[INFO|2026-09-12 14:11:41] logging.py:144 >> {'loss': 1.3282, 'learning_rate':
[INFO|2026-09-12 14:12:16] logging.py:144 >> {'loss': 0.6698, 'learning_rate':
[INFO|2026-09-12 14:12:51] logging.py:144 >> {'loss': 0.4298, 'learning_rate':
[INFO|2026-09-12 14:13:21] logging.py:144 >> {'loss': 0.3445, 'learning_rate':
[INFO|2026-09-12 14:13:55] logging.py:144 >> {'loss': 0.3147, 'learning_rate':
[INFO|2026-09-12 14:14:29] logging.py:144 >> {'loss': 0.2758, 'learning_rate':
[INFO|2026-09-12 14:15:04] logging.py:144 >> {'loss': 0.2693, 'learning_rate':
[INFO|2026-09-12 14:15:38] logging.py:144 >> {'loss': 0.2027, 'learning_rate':
[INFO|2026-09-12 14:16:12] logging.py:144 >> {'loss': 0.2061, 'learning_rate':
[INFO|2026-09-12 14:16:41] logging.py:144 >> {'loss': 0.2405, 'learning_rate':
[INFO|2026-09-12 14:17:16] logging.py:144 >> {'loss': 0.1919, 'learning_rate':
[INFO|2026-09-12 14:17:52] logging.py:144 >> {'loss': 0.1977, 'learning_rate':
[INFO|2026-09-12 14:18:26] logging.py:144 >> {'loss': 0.1324, 'learning_rate':
[INFO|2026-09-12 14:19:01] logging.py:144 >> {'loss': 0.2562, 'learning_rate':
[INFO|2026-09-12 14:19:34] logging.py:144 >> {'loss': 0.1914, 'learning_rate':
[INFO|2026-09-12 14:20:02] logging.py:144 >> {'loss': 0.2072, 'learning_rate':
[INFO|2026-09-12 14:20:02] trainer.py:3815 >> Saving model checkpoint to saves
[INFO|2026-09-12 14:20:03] configuration_utils.py:780 >> loading configuration
[INFO|2026-09-12 14:20:03] configuration_utils.py:856 >> Model config Qwen3Con
    "architectures": [
        "Qwen3ForCausalLM"
    ],
    "attention_bias": false,
    "attention_dropout": 0.0,
    "bos_token_id": 151643,
    "dtype": "bfloat16",
    "eos_token_id": 151643,

```


Training completed. Do not forget to share your model on huggingface.co/models

[illegible]

```

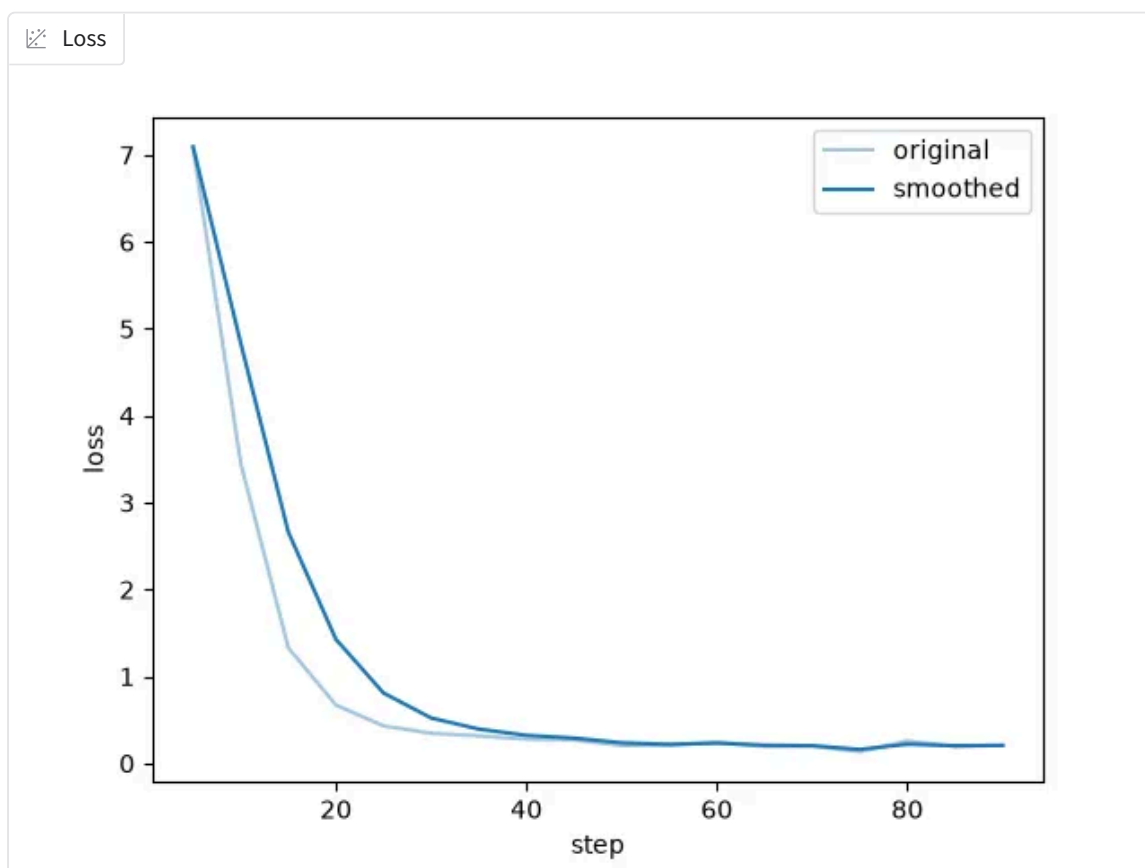
"rope_parameters": {
  "rope_theta": 1000000,
  "rope_type": "default"
},
"sliding_window": null,
"tie_word_embeddings": true,
"transformers_version": "5.8.0",
"use_cache": true,
"use_sliding_window": false,
"vocab_size": 151936
}

```

```

[INFO|2026-09-12 14:20:04] tokenization_utils_base.py:3281 >> chat template sa
[INFO|2026-09-12 14:20:04] tokenization_utils_base.py:2114 >> tokenizer config

```



Device memory

Current memory usage of the device (GB).

28.08



0



28.08